Examples of maximal function fields

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Abstract

In this note, we give examples of maximal function fields of genus g over a finite field with q^2 elements for certain small values of q and g. The examples were obtained using AI tools. The function fields are given by explicit equations and we provide a Magma script that can be used to compute the relevant invariants, i.e., the full constant field, genus and number of rational places. Each example improves a lower bound recorded in the Tables of Curves with Many Points [3] as of November 6, 2026. The note is intended to be extended as further examples are found.

1 Introduction

Let q be a prime power. An algebraic function field of genus g with full constant field $\mathbb{F}_{q^2}$ is maximal if it has

$$q^2 + 1 + 2qg$$

rational places. Here a rational place is a place of degree one over the full constant field. Rational places correspond to rational points on the associated smooth projective curve. The examples in this note improve the corresponding lower bounds recorded in the Tables of Curves with Many Points [3] as of November 6th 2026.

In each example, x is transcendental over the base field, and the displayed equations define the remaining algebraic generators. The associated curve is the non-singular projective model corresponding to the function field.

Each polynomial used to specify an element a below is irreducible over the prime field. Its roots generate the indicated finite field and are Frobenius conjugate, so any root gives the stated genus and number of rational places.

Appendix A contains the Magma [1] code used for computing the stated invariants.

2 Examples

Example 1 ($q = 13$, $g = 7$)**.** Consider the function field $\mathbb{F}_{13^2}(x, y, z)$ defined by

$$y^2 = 11x(x^2 - 6)(x^2 + 9x + 6), \qquad z^2 = x(x^2 + 4)(y + x^2 - x + 7).$$

A direct computation using Magma shows that its full constant field is $\mathbb{F}_{13^2}$, its genus is 7, and it has 352 rational places. It is therefore maximal.

Remark. The subfield $\mathbb{F}_{13^2}(x, y)$ has genus 2 and over the algebraic closure $\overline{\mathbb{F}}_{13^2}$ it is isomorphic to a function field that was studied by Ibukiyama–Katsura–Oort in [5]. In fact,

choose $a \in \overline{\mathbb{F}}_{13^2}$ with $a^4 = 5$ and define

$$u = \frac{7 + 10a^2x}{6 + 10a^2x}, \qquad v = \frac{(7 + 3a^2)y}{a(6 + 10a^2x)^3}.$$

Then $\overline{\mathbb{F}}_{13^2}(x, y) = \overline{\mathbb{F}}_{13^2}(u, v)$ and

$$v^2 = u(u^2 - 1)(u^2 - 8 - 7a^2),$$

so the function field is of type (b) in [5, Section 1.3].

Example 2 ($q = 13$, $g = 8$)**.** Let a be an element of $\mathbb{F}_{13^2}$ such that $a^2 = 5$. Consider the function field $\mathbb{F}_{13^2}(x, y, z)$ defined by

$$y^2 = (x^2 - 1)(x^4 + (3a + 5)x^2 + 1),$$
$$z^3 = \frac{(7a + 7)x^4 + (3a + 8)x^2 + 1 - a + \big((11a + 5)x^2 + a + 8\big)y}{x^2 + 9a + 2}.$$

A direct computation using Magma shows that its full constant field is $\mathbb{F}_{13^2}$, its genus is 8, and it has 378 rational places. It is therefore maximal.

Remark. Choose $b \in \overline{\mathbb{F}}_{13^2}$ with $b^2 = a$, and define

$$u = \frac{(11a + 5)(1 + x)}{1 - x}, \qquad v = \frac{(7 + 4a)y}{b(1 - x)^3}.$$

Then $\overline{\mathbb{F}}_{13^2}(x, y) = \overline{\mathbb{F}}_{13^2}(u, v)$ with $v^2 = u(u^2 - 1)(u^2 - 8 - 7a)$, i.e., it is the same genus two function field from [5] as in Example 1.

Example 3 ($q = 19$, $g = 10$)**.** Consider the function field $\mathbb{F}_{19^2}(x, y, z)$ defined by

$$y^2 = (x^2 - 1)(x^2 - 4)(x^2 + 2),$$
$$z^2 = x(x + 1)(x - 2)(x + 6)(x^2 - 6)(x^2 - 9)(x^2 + 3).$$

A direct computation using Magma shows that its full constant field is $\mathbb{F}_{19^2}$, its genus is 10, and it has 742 rational places. It is therefore maximal.

Remark. The subfield $\mathbb{F}_{19^2}(x, y)$ has genus 2 and also belongs to the family studied in [5, Section 1.3]. In fact, with

$$u = \frac{x - 6}{2(x + 6)}, \qquad v = \frac{6y}{(x + 6)^3},$$

we have $\mathbb{F}_{19^2}(x, y) = \mathbb{F}_{19^2}(u, v)$ and $v^2 = u(u^2 - 1)(u^2 - 6)$. Moreover, the subfield $\mathbb{F}_{19^2}(x, z)$ has genus 4 and it is a subfield of the Hermitian function field. In fact, choose $b \in \mathbb{F}_{19^2}$ with $b^2 = -1$ and put

$$s = \frac{x + 10 - 4b}{x + 10 + 4b}, \qquad t = \frac{6z}{(x + 10 + 4b)^5}.$$

Then $\mathbb{F}_{19^2}(x, z) = \mathbb{F}_{19^2}(s, t)$ with $t^2 + (s^{10} + 1) = 0$, and this is a subfield of the Hermitian function field (see [2, Example 6.4]).

Example 4 ($q = 19$, $g = 11$)**.** Consider the function field $\mathbb{F}_{19^2}(x, y, z)$ defined by

$$y^2 = \prod_{i\in\{4,6,9,11,17\}} (x^2 - i), \qquad z^2 = x(x^2 - 4)(x^2 - 5)(x^2 - 11).$$

A direct computation using Magma shows that its full constant field is $\mathbb{F}_{19^2}$, its genus is 11, and it has 780 rational places. It is therefore maximal.

Remark. The subfield $\mathbb{F}_{19^2}(x, y)$ is isomorphic to the genus four subfield in the previous example. In fact, choose $b \in \mathbb{F}_{19^2}$ with $b^2 = -1$ and define

$$s = \frac{x - 2b}{x + 2b}, \qquad t = \frac{6y}{(x + 2b)^5}.$$

Then $\mathbb{F}_{19^2}(x, y) = \mathbb{F}_{19^2}(s, t)$ with $t^2 + (s^{10} + 1) = 0$ (see [2, Example 6.4]). Moreover, the subfield $\mathbb{F}_{19^2}(x, z)$ has genus 3 and it belongs to a family of hyperelliptic function fields that was studied in [6]. In fact, defining

$$u = -b\,\frac{x - 9}{x + 9}, \qquad v = \frac{4(1 - b)z}{(x + 9)^4},$$

we get $\mathbb{F}_{19^2}(x, z) = \mathbb{F}_{19^2}(u, v)$ and $v^2 = u^7 + u$, i.e., it is a function field of the type described in [6, Proposition 4].

Example 5 ($q = 5^2$, $g = 5$)**.** Let a be an element of $\mathbb{F}_{5^4}$ such that $a^4 + 4a^2 + 4a + 2 = 0$. Consider the function field $\mathbb{F}_{5^4}(x, y, z)$ defined by

$$\begin{aligned} y^2 &= x^3 + a^3, \\ z^3 &= x^3 - x^2 + (4a^3 + 2a^2 + 4a + 1)x + 3a^3 + 4a^2 \\ &\quad + (2ax + 4a^3 + a + 4)y. \end{aligned}$$

A direct computation using Magma shows that its full constant field is $\mathbb{F}_{5^4}$, its genus is 5, and it has 876 rational places. It is therefore maximal.

Remark. The elliptic subfield $\mathbb{F}_{5^4}(x, y)$ is well known to be maximal; see [3].

Example 6 ($q = 5^2$, $g = 7$)**.** Let a be an element of $\mathbb{F}_{5^4}$ such that $a^4 + 4a^2 + 4a + 2 = 0$. Choose a primitive third root of unity $\zeta \in \mathbb{F}_{5^4}$ and put

$$\begin{aligned} f(x, y) = {} & (a^2 + 4a + 3)x^3 + (a^3 + a^2 + 2a + 4)x^2 \\ & + (3a^3 + 2a^2 + 3)x + a^3 + a^2 + a + 4 \\ & + (x + 3a^2 + 4a + 2)y. \end{aligned}$$

Consider the function field $\mathbb{F}_{5^4}(x, y, z)$ defined by

$$y^2 = x^3 + a^3,$$

$$\left(z^2 - f(x, y) - f(\zeta x, y)\right)^2 = 4f(x, y)f(\zeta x, y).$$

A direct computation using Magma shows that its full constant field is $\mathbb{F}_{5^4}$, its genus is 7, and it has 976 rational places. It is therefore maximal.

Remark. The elliptic subfield $\mathbb{F}_{5^4}(x,y)$ is well known to be maximal; see [3].

Example 7 ($q=7^2$, $g=5$)**.** Choose $a\in\mathbb{F}_{7^2}\subset\mathbb{F}_{7^4}$ with $a^2=-1$, and let $\nu\in\mathbb{F}_{7^4}$ be any nonsquare. Consider the function field $\mathbb{F}_{7^4}(x,y)$ defined by

$$y^2=\nu(x^6-1)(x^6+ax^3+1).$$

A direct computation using Magma shows that its full constant field is $\mathbb{F}_{7^4}$, its genus is 5, and it has 2892 rational places. It is therefore maximal.

Remark. Replacing y by y/b, where $b^2=\nu$ in $\overline{\mathbb{F}}_7$, shows that the function field $\overline{\mathbb{F}}_7(x,y)$ belongs to the family studied in characteristic zero by Hidalgo–Shaska [4, Table 3, genus-five Nr 15].

Example 8 ($q=13^2$, $g=5$)**.** Let a be an element of $\mathbb{F}_{13^4}$ such that $a^4=5$. Consider the function field $\mathbb{F}_{13^4}(x,y)$ defined by

$$y^2=a\big(x^{12}+(a^2+3)x^9+(8-2a^2)x^6+1\big).$$

A direct computation using Magma shows that its full constant field is $\mathbb{F}_{13^4}$, its genus is 5, and it has 30252 rational places. It is therefore maximal.

Remark. Replacing y by y/b, where $b^2=a$ in $\overline{\mathbb{F}}_{13}$, shows that the function field $\overline{\mathbb{F}}_{13}(x,y)$ belongs to the family studied in characteristic zero by Hidalgo–Shaska [4, Table 3, genus-five Nr 2].

Example 9 ($q=17^2$, $g=7$)**.** Let a be an element of $\mathbb{F}_{17^4}$ such that $a^4+7a^2+10a+3=0$. Choose a primitive third root of unity $\zeta\in\mathbb{F}_{17^4}$ and put

$$\begin{aligned}f(x,y)=\;&(10a^3+9a^2+11a+3)x^3\\&+(6a^3+12a^2+10a+11)x^2\\&+(15a^3+12a^2+4a+5)x+a^3+10a^2+3a+9\\&+(x+5a^3+a^2+2a+10)y.\end{aligned}$$

Consider the function field $\mathbb{F}_{17^4}(x,y,z)$ defined by

$$y^2=x^3+a^3,$$

$$\big(z^2-f(x,y)-f(\zeta x,y)\big)^2=4f(x,y)f(\zeta x,y).$$

A direct computation using Magma shows that its full constant field is $\mathbb{F}_{17^4}$, its genus is 7, and it has 87568 rational places. It is therefore maximal.

Remark. The elliptic subfield $\mathbb{F}_{17^4}(x,y)$ is well known to be maximal; see [3].

Appendix A: Magma code

The following Magma programs construct the function fields from Section 2 and compute their full constant fields, genera and numbers of rational places.

Example 1

```
q := 13;
k := GF(q^2);

K<x> := FunctionField(k);
P<T> := PolynomialRing(K);
E<y> := ext<K | T^2 - 11*x*(x^2-6)*(x^2+9*x+6)>;
P<T> := PolynomialRing(E);
F<z> := ext<E | T^2 - x*(x^2+4)*(y+x^2-x+7)>;

// Compute invariants of the function field.
c := #ExactConstantField(F);
g := Genus(F);
N := NumberOfPlacesOfDegreeOneECF(F);
maximal := c eq q^2 and N eq q^2 + 1 + 2*q*g;

printf "Full constant field size: %o (expected %o)\n", c, q^2;
printf "Genus: %o (expected 7)\n", g;
printf "Rational places: %o (expected 352)\n", N;
printf "Maximal over GF(%o): %o\n", q^2, maximal;
```

Example 2

```
q := 13;
p := Characteristic(GF(q));
P<t> := PolynomialRing(GF(p));
k<a> := ext<GF(p) | t^2-5>;

K<x> := FunctionField(k);
P<T> := PolynomialRing(K);
E<y> := ext<K | T^2-(x^2-1)*(x^4+(3*a+5)*x^2+1)>;
h := ((7*a+7)*x^4+(3*a+8)*x^2+1-a
      +((11*a+5)*x^2+a+8)*y)/(x^2+9*a+2);
P<T> := PolynomialRing(E);
F<z> := ext<E | T^3-h>;

// Compute invariants of the function field.
c := #ExactConstantField(F);
g := Genus(F);
N := NumberOfPlacesOfDegreeOneECF(F);
maximal := c eq q^2 and N eq q^2 + 1 + 2*q*g;

printf "Full constant field size: %o (expected %o)\n", c, q^2;
printf "Genus: %o (expected 8)\n", g;
printf "Rational places: %o (expected 378)\n", N;
printf "Maximal over GF(%o): %o\n", q^2, maximal;
```

Example 3

```
q := 19;
k := GF(q^2);

K<x> := FunctionField(k);
A := (x^2-1)*(x^2-4)*(x^2+2);
B := x*(x+1)*(x-2)*(x+6)*(x^2-6)*(x^2-9)*(x^2+3);
P<T> := PolynomialRing(K);
E<y> := ext<K | T^2-A>;
P<T> := PolynomialRing(E);
F<z> := ext<E | T^2-B>;

// Compute invariants of the function field.
c := #ExactConstantField(F);
g := Genus(F);
N := NumberOfPlacesOfDegreeOneECF(F);
maximal := c eq q^2 and N eq q^2 + 1 + 2*q*g;

printf "Full constant field size: %o (expected %o)\n", c, q^2;
printf "Genus: %o (expected 10)\n", g;
printf "Rational places: %o (expected 742)\n", N;
printf "Maximal over GF(%o): %o\n", q^2, maximal;
```

Example 4

```
q := 19;
k := GF(q^2);

K<x> := FunctionField(k);
A := &*[x^2-i : i in [4,6,9,11,17]];
B := x*(x^2-4)*(x^2-5)*(x^2-11);
P<T> := PolynomialRing(K);
E<y> := ext<K | T^2-A>;
P<T> := PolynomialRing(E);
F<z> := ext<E | T^2-B>;

// Compute invariants of the function field.
c := #ExactConstantField(F);
g := Genus(F);
N := NumberOfPlacesOfDegreeOneECF(F);
maximal := c eq q^2 and N eq q^2 + 1 + 2*q*g;

printf "Full constant field size: %o (expected %o)\n", c, q^2;
printf "Genus: %o (expected 11)\n", g;
printf "Rational places: %o (expected 780)\n", N;
printf "Maximal over GF(%o): %o\n", q^2, maximal;
```

Example 5

```
q := 25;
p := Characteristic(GF(q));
P<t> := PolynomialRing(GF(p));
k<a> := ext<GF(p) | t^4+4*t^2+4*t+2>;

K<x> := FunctionField(k);
P<T> := PolynomialRing(K);
E<y> := ext<K | T^2-x^3-a^3>;
h := x^3-x^2+(4*a^3+2*a^2+4*a+1)*x+3*a^3+4*a^2
     +(2*a*x+4*a^3+a+4)*y;
P<T> := PolynomialRing(E);
F<z> := ext<E | T^3-h>;

// Compute invariants of the function field.
c := #ExactConstantField(F);
g := Genus(F);
N := NumberOfPlacesOfDegreeOneECF(F);
maximal := c eq q^2 and N eq q^2 + 1 + 2*q*g;

printf "Full constant field size: %o (expected %o)\n", c, q^2;
printf "Genus: %o (expected 5)\n", g;
printf "Rational places: %o (expected 876)\n", N;
printf "Maximal over GF(%o): %o\n", q^2, maximal;
```

Example 6

```
q := 25;
p := Characteristic(GF(q));
P<t> := PolynomialRing(GF(p));
k<a> := ext<GF(p) | t^4+4*t^2+4*t+2>;
P<t> := PolynomialRing(k);
zeta := Roots(t^2+t+1)[1][1];

K<x> := FunctionField(k);
P<T> := PolynomialRing(K);
E<y> := ext<K | T^2-x^3-a^3>;

// Evaluate f at x and at zeta*x.
function f(x,y)
    return (a^2+4*a+3)*x^3+(a^3+a^2+2*a+4)*x^2
           +(3*a^3+2*a^2+3)*x+a^3+a^2+a+4
           +(x+3*a^2+4*a+2)*y;
end function;

f0 := f(x,y);
f1 := f(zeta*x,y);
P<T> := PolynomialRing(E);
F<z> := ext<E | (T^2-f0-f1)^2-4*f0*f1>;

// Compute invariants of the function field.
c := #ExactConstantField(F);
g := Genus(F);
N := NumberOfPlacesOfDegreeOneECF(F);
maximal := c eq q^2 and N eq q^2 + 1 + 2*q*g;

printf "Full constant field size: %o (expected %o)\n", c, q^2;
printf "Genus: %o (expected 7)\n", g;
printf "Rational places: %o (expected 976)\n", N;
printf "Maximal over GF(%o): %o\n", q^2, maximal;
```

Example 7

```
q := 49;
p := Characteristic(GF(q));
P<t> := PolynomialRing(GF(p));
k<b> := ext<GF(p) | t^4+5*t^2+5>;
a := 3*b^2+4;
nu := b;
// Here a^2=-1 and nu is a nonsquare since nu^((q^2-1) div 2) = -1.

K<x> := FunctionField(k);
P<T> := PolynomialRing(K);
F<y> := ext<K | T^2-nu*(x^6-1)*(x^6+a*x^3+1)>;

// Compute invariants of the function field.
c := #ExactConstantField(F);
g := Genus(F);
N := NumberOfPlacesOfDegreeOneECF(F);
maximal := c eq q^2 and N eq q^2 + 1 + 2*q*g;

printf "Full constant field size: %o (expected %o)\n", c, q^2;
printf "Genus: %o (expected 5)\n", g;
printf "Rational places: %o (expected 2892)\n", N;
printf "Maximal over GF(%o): %o\n", q^2, maximal;
```

Example 8

```
q := 169;
p := Characteristic(GF(q));
P<t> := PolynomialRing(GF(p));
k<a> := ext<GF(p) | t^4-5>;

K<x> := FunctionField(k);
P<T> := PolynomialRing(K);
F<y> := ext<K | T^2-a*(x^12+(a^2+3)*x^9+(8-2*a^2)*x^6+1)>;

// Compute invariants of the function field.
c := #ExactConstantField(F);
g := Genus(F);
N := NumberOfPlacesOfDegreeOneECF(F);
maximal := c eq q^2 and N eq q^2 + 1 + 2*q*g;

printf "Full constant field size: %o (expected %o)\n", c, q^2;
printf "Genus: %o (expected 5)\n", g;
printf "Rational places: %o (expected 30252)\n", N;
printf "Maximal over GF(%o): %o\n", q^2, maximal;
```

Example 9

```
q := 289;
p := Characteristic(GF(q));
P<t> := PolynomialRing(GF(p));
k<a> := ext<GF(p) | t^4+7*t^2+10*t+3>;
P<t> := PolynomialRing(k);
zeta := Roots(t^2+t+1)[1][1];

function f(x,y)
    return (10*a^3+9*a^2+11*a+3)*x^3
           +(6*a^3+12*a^2+10*a+11)*x^2
           +(15*a^3+12*a^2+4*a+5)*x+a^3+10*a^2+3*a+9
           +(x+5*a^3+a^2+2*a+10)*y;
end function;

K<x> := FunctionField(k);
P<T> := PolynomialRing(K);
E<y> := ext<K | T^2-x^3-a^3>;

f0 := f(x,y);
f1 := f(zeta*x,y);
P<T> := PolynomialRing(E);
F<z> := ext<E | (T^2-f0-f1)^2-4*f0*f1>;

// Compute invariants of the function field.
c := #ExactConstantField(F);
g := Genus(F);
N := NumberOfPlacesOfDegreeOneECF(F);
maximal := c eq q^2 and N eq q^2 + 1 + 2*q*g;

printf "Full constant field size: %o (expected %o)\n", c, q^2;
printf "Genus: %o (expected 7)\n", g;
printf "Rational places: %o (expected 87568)\n", N;
printf "Maximal over GF(%o): %o\n", q^2, maximal;
```

AI-use declaration

The AI tools ChatGPT and Codex (OpenAI) and Claude (Anthropic) were used in the search for examples, algebraic calculations, development of Magma programs and preparation of this note.

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